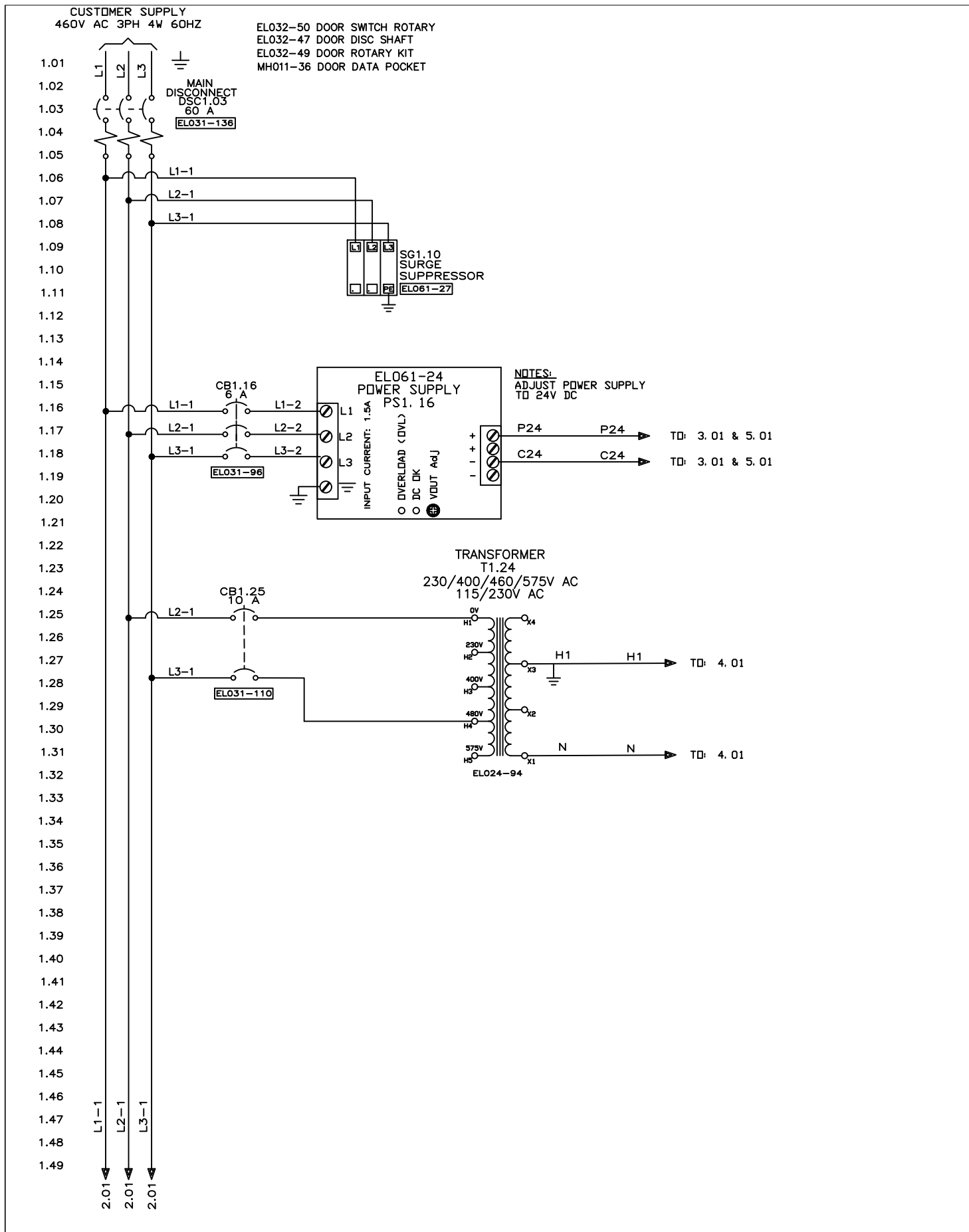
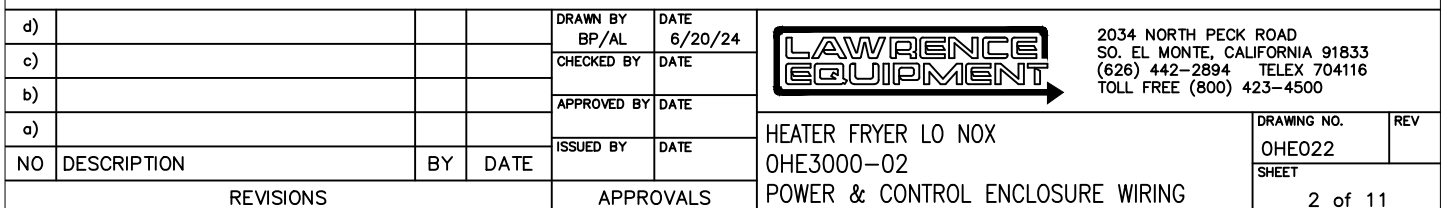
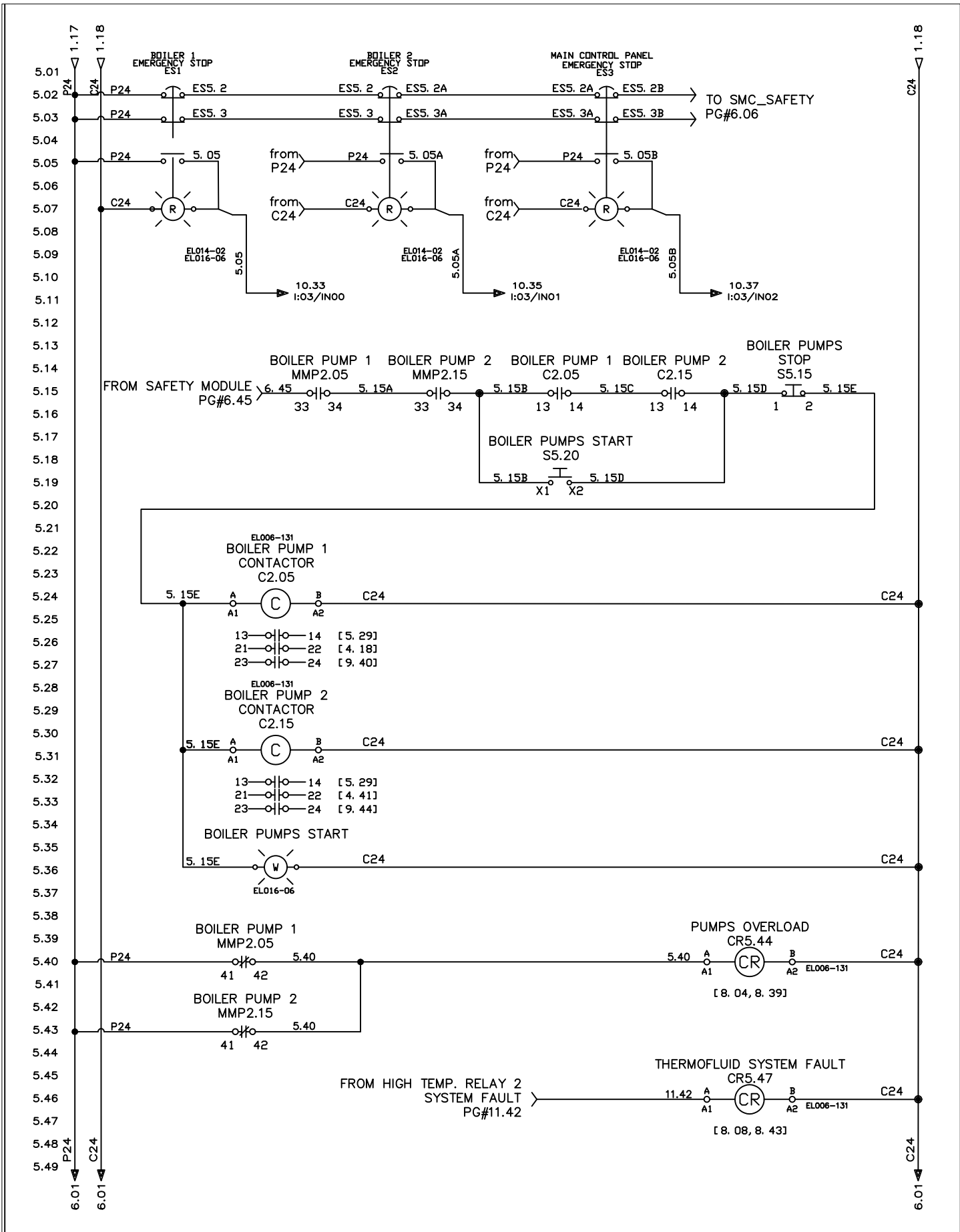


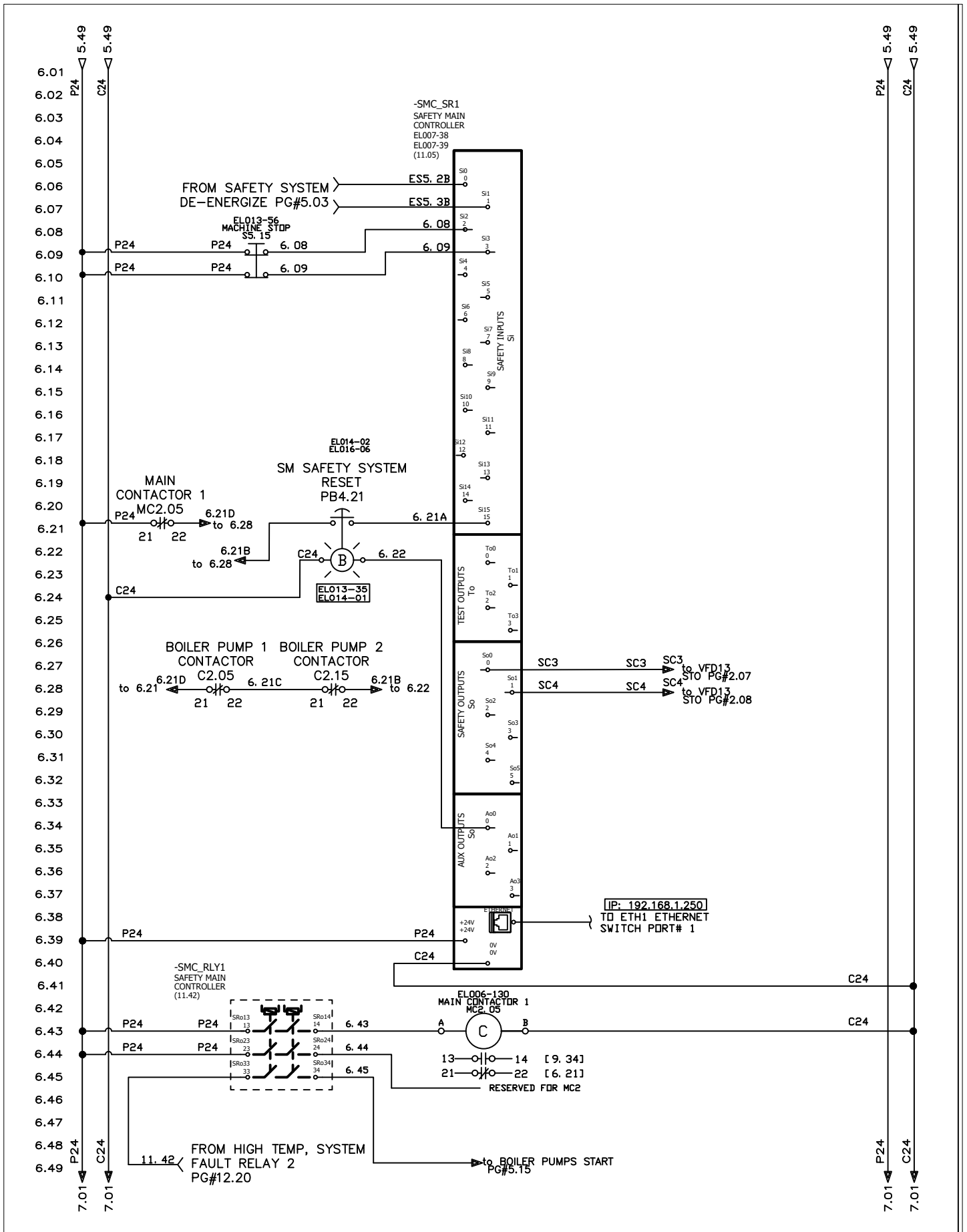


d)				DRAWN BY BP/AL	DATE 6/20/24		2034 NORTH PECK ROAD SO. EL MONTE, CALIFORNIA 91833 (626) 442-2894 TELEX 704116 TOLL FREE (800) 423-4500		
c)				CHECKED BY	DATE				
b)				APPROVED BY	DATE				
a)				ISSUED BY	DATE				
NO	DESCRIPTION		BY	DATE	HEATER FRYER LO NOX OHE3000-02 POWER & CONTROL ENCLOSURE WIRING			DRAWING NO. OHE022	REV
REVISIONS				APPROVALS				SHEET	
								1 of 11	

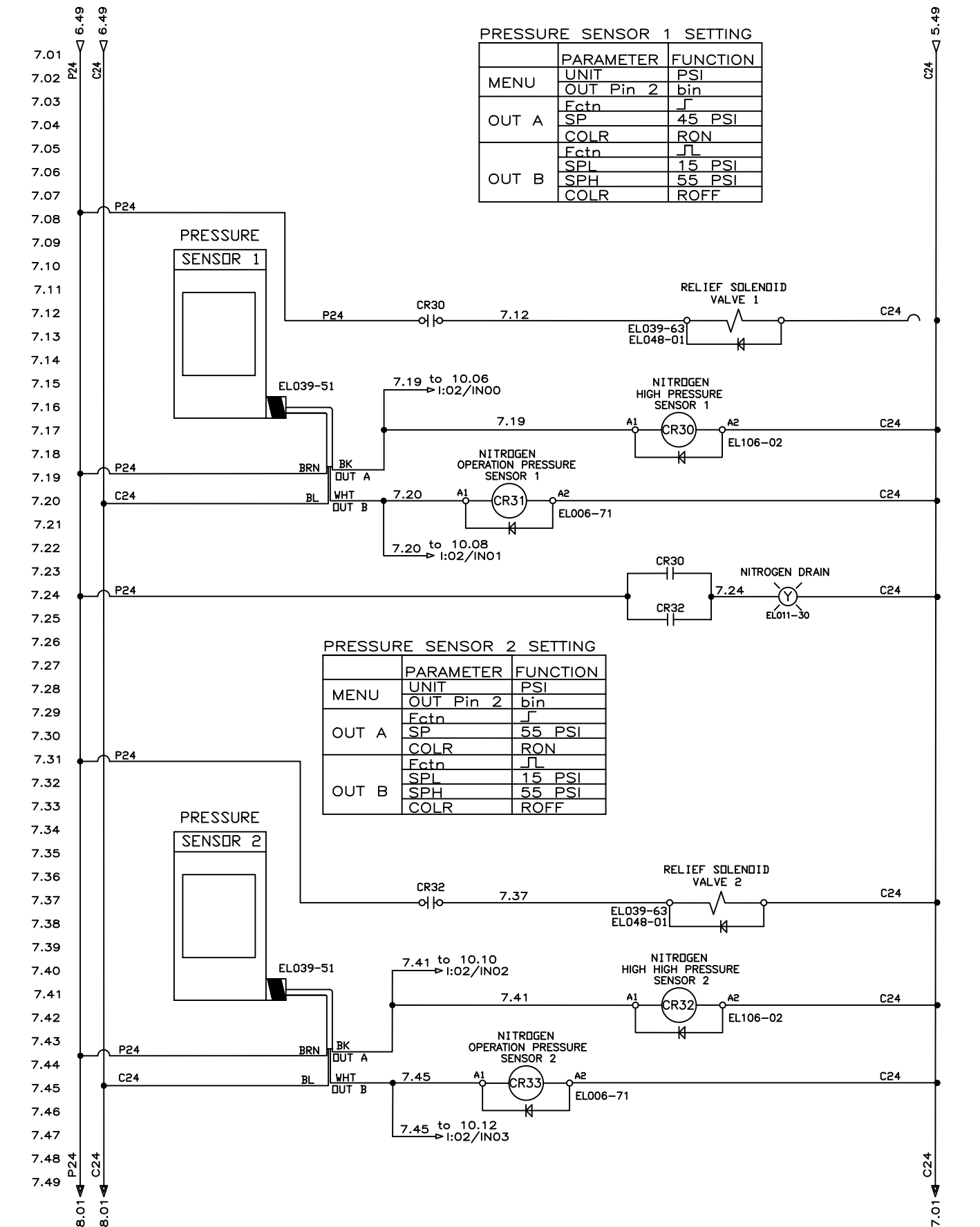




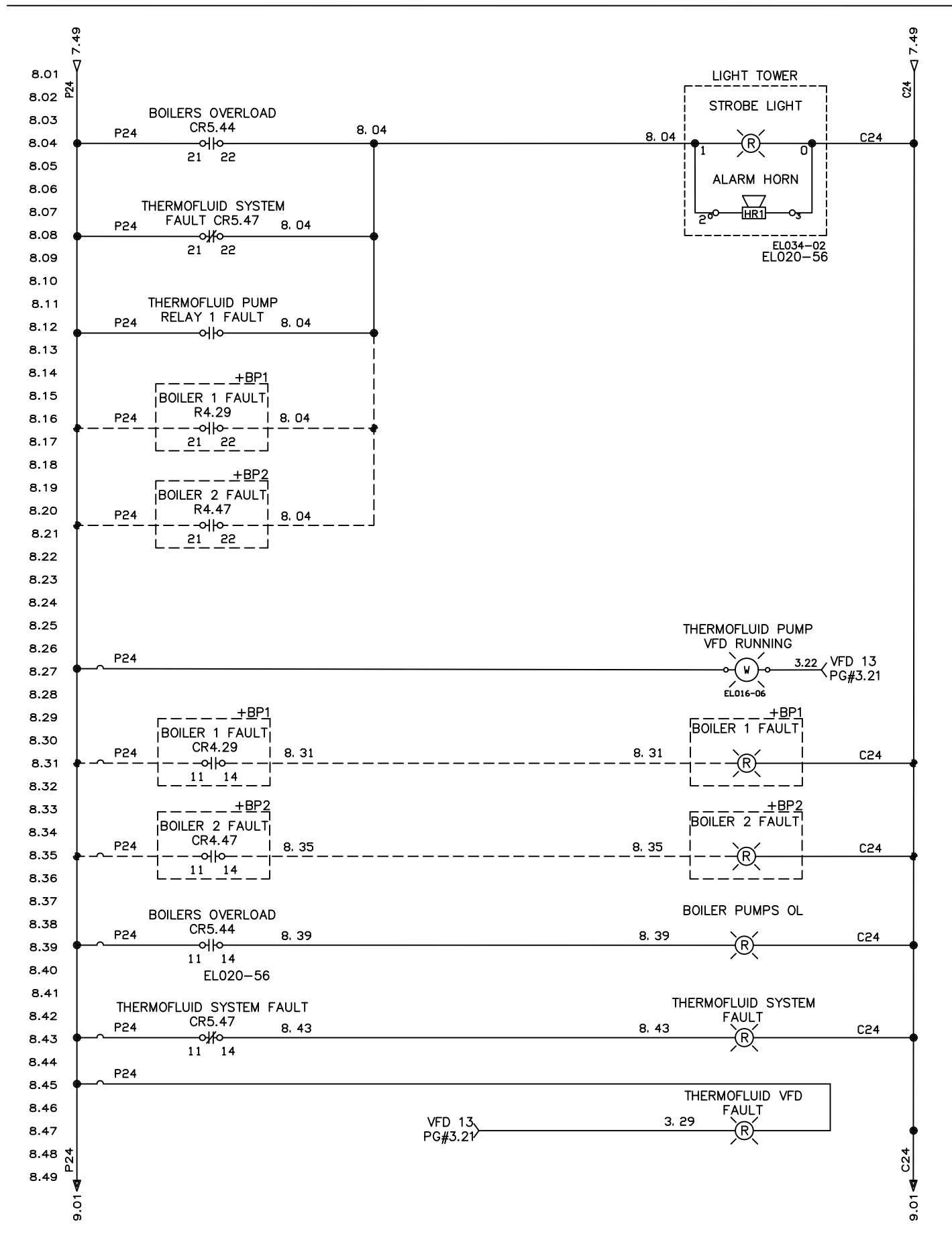
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REVISIONS				APPROVALS		SHEET			
						5 of 11			



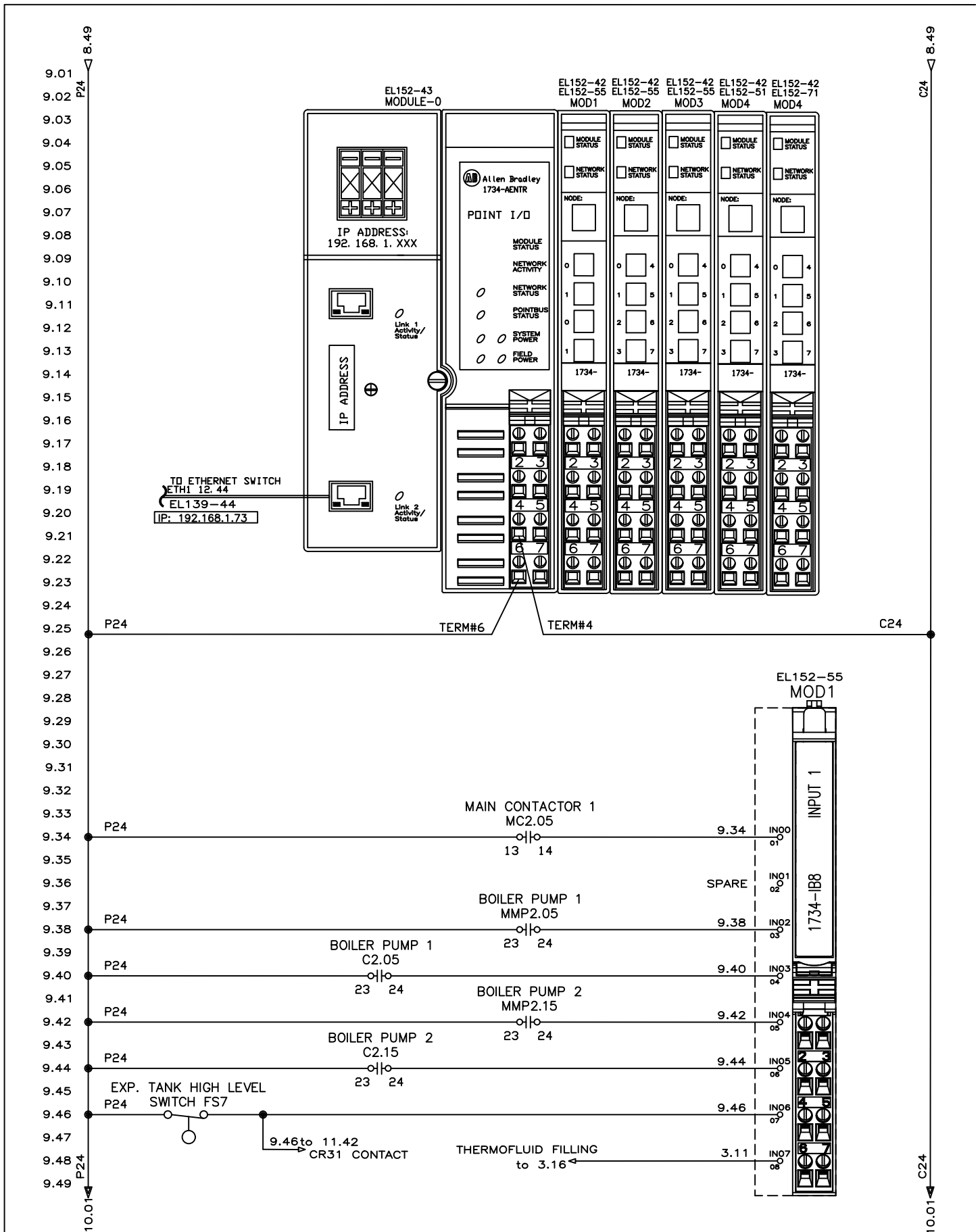
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d)				DRAWN BY	DATE		2034 NORTH PECK ROAD SO. EL MONTE, CALIFORNIA 91833 (626) 442-2894 TELEX 704116 TOLL FREE (800) 423-4500	
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						7 of 11		



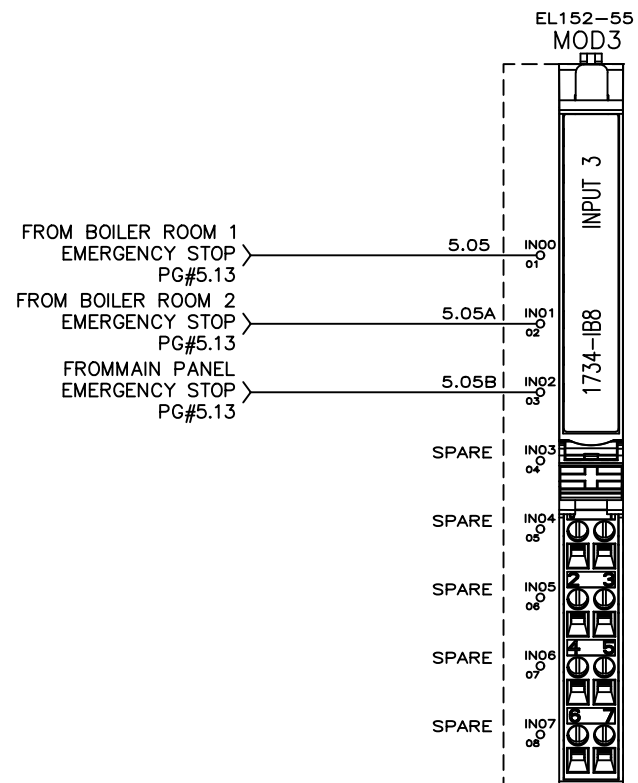
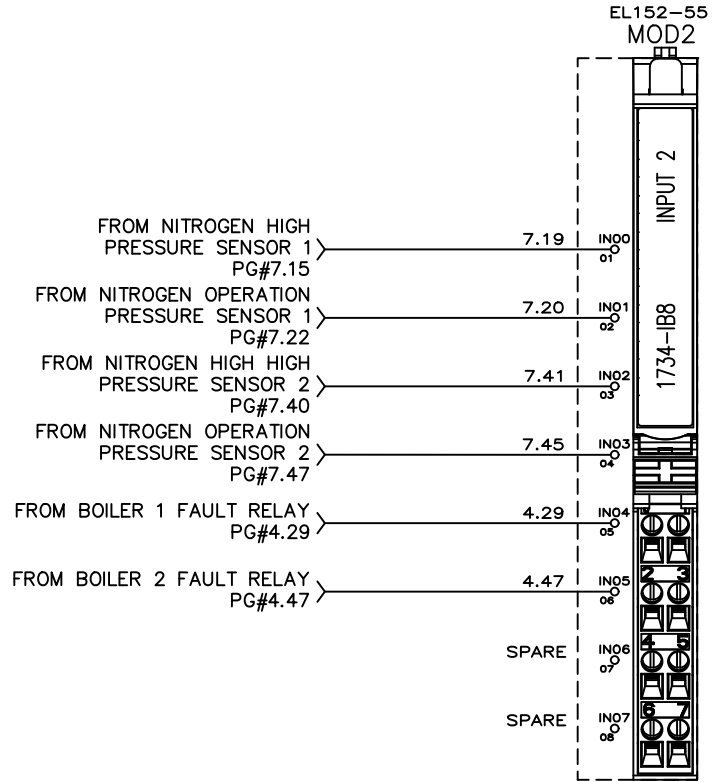
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REVISIONS										



d)				DRAWN BY BP/AL	DATE 6/20/24		2034 NORTH PECK ROAD SO. EL MONTE, CALIFORNIA 91833 (626) 442-2894 TELEX 704116 TOLL FREE (800) 423-4500	
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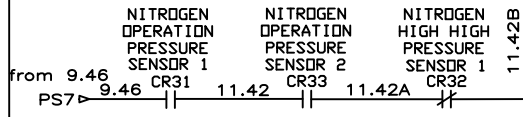
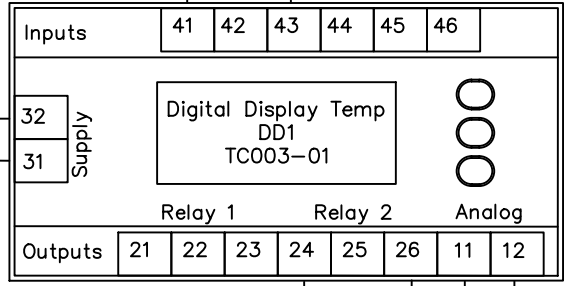
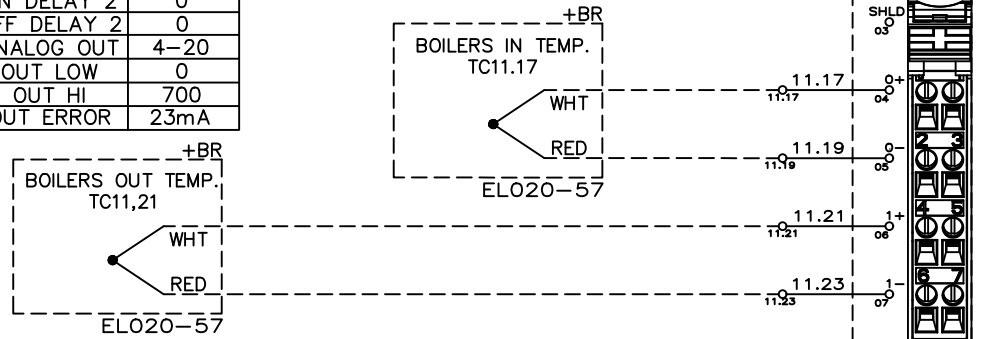


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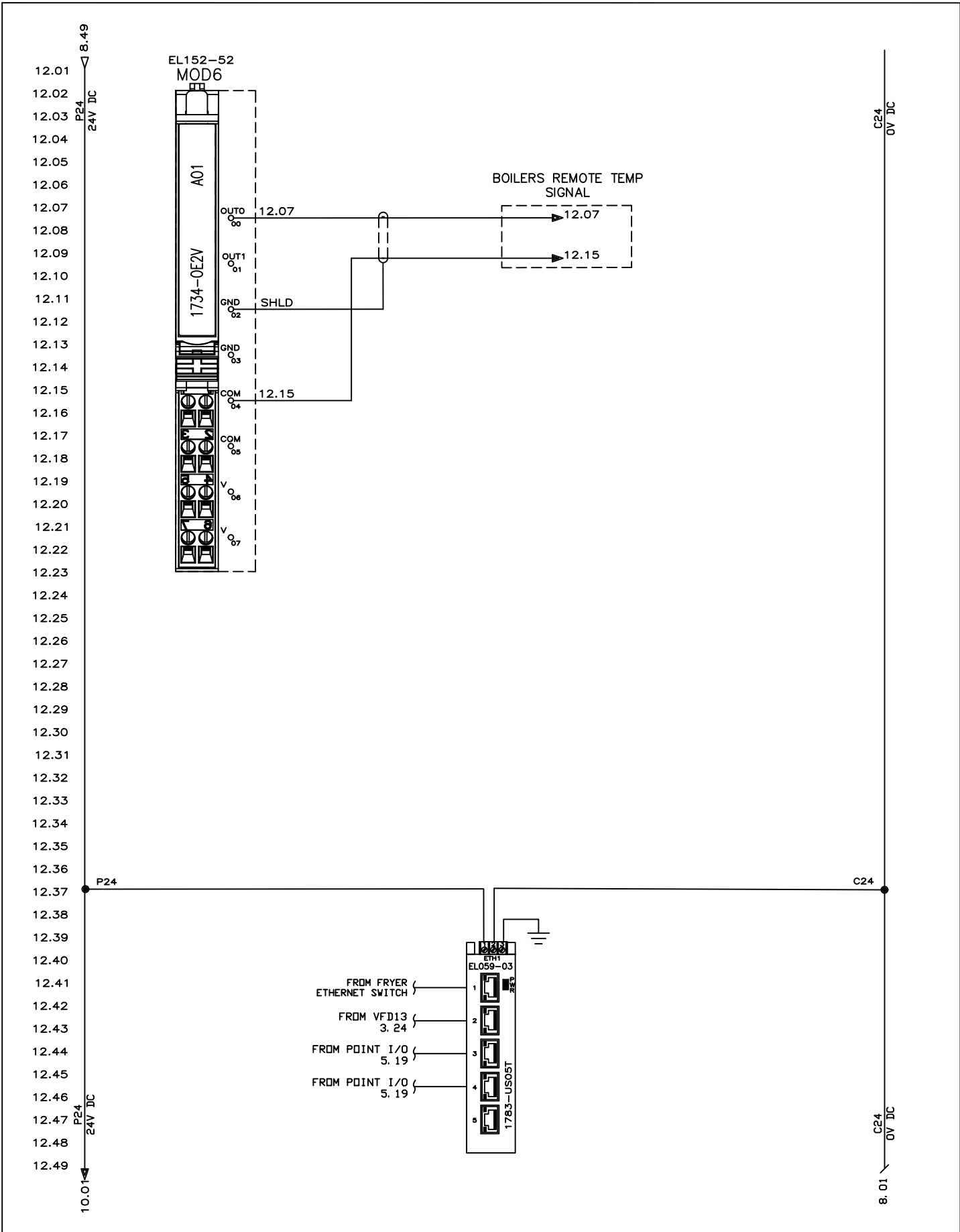
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Digital Display Settings		
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SENSOR TYPE	TC	
TC TYPE	TC J	
DEC. POINT	111.1	
UNIT	°F	
RELAY 1	SET	
SET POINT 1	195	
HYST 1	5.0	
ACT. 1	INCR	
ERR. 1	ACTIV	
ON DELAY 1	0	
OFF DELAY 1	0	
RELAY 2	SET	
SET POINT 2	195	
HYST 2	5.0	
ACT. 2	INCR	
ERR. 2	ACTIV	
ON DELAY 2	0	
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ANALOG OUT	4-20	
OUT LOW	0	
OUT HI	700	
OUT ERROR	23mA	



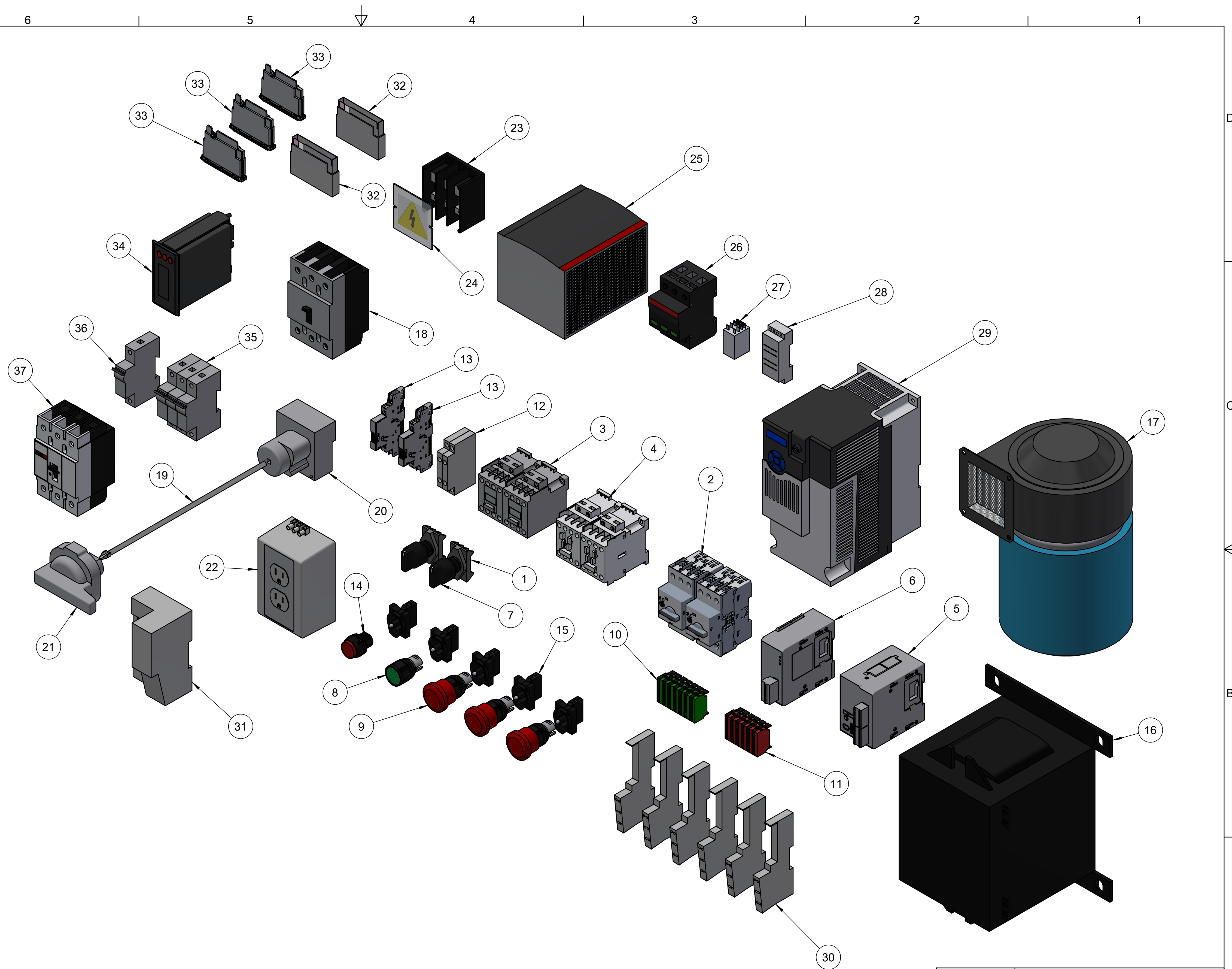
TO THERMOFLUID SYSTEM
FAULT RELAY
PG#5.47
11.42

TO SMC SAFETY MODULE
SMC RELAY OUTPUT 33
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d)				DRAWN BY BP/AL	DATE 6/20/24	 2034 NORTH PECK ROAD SO. EL MONTE, CALIFORNIA 91833 (626) 442-2894 TELEX 704116 TOLL FREE (800) 423-4500	DRAWING NO. OHE022 SHEET	REV 11 of 11	
c)				CHECKED BY	DATE				
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NO	DESCRIPTION		BY	DATE	HEATER FRYER LO NOX OHE3000-02 POWER & CONTROL ENCLOSURE WIRING				
REVISIONS				APPROVALS					

8		7		
MATERIALS LIST				
ITEM	LE PART NUMBER	DESCRIPTION	QTY	
D	1	EH999-18	BASE FLANGE FOR 3 CONTACTS	2
	2	EL006-169	MANUAL MTR STARTER 6.30-10.00	2
	3	EL006-97	CONTACTOR 9AMP 24VDC COIL 460V	2
	4	EL006-151	CONTACTOR 23 AMP 24 VDC COIL	2
	5	EL007-38	SAFTEY MAIN CONTROLLER STD	1
	6	EL007-39	SAFETY RELAY OUTPUT UNIT	1
	7	EL012-11	SELECTOR SWITCH 2POS KNOB	2
	8	EL013-31	PUSH BOTTON FLUSH GREEN 22.5MM	1
	9	EL013-34	PUSH BUTTON ILLUM MUSHROOM RED	3
	10	EL014-01	CONTACT BLOCK N.O.	7
C	11	EL014-02	CONTACT BLOCK N.C.	6
	12	EL014-32	CONTACT BLOCKS AUX 1NO 1NC	2
	13	EL014-67	CONTACT BLOCKS AUX 1NO 1NC	2
	14	EL013-45	PUSH BUTTON EXTEND RED 22MM	1
	15	EL016-06	LIGHT TERM W/BULB LATCH	5
	16	EL024-94	TRANS 5.0KVA 400-460V/115 230V	1
	17	EL027-08	CABINET COOLER 25 CUBIC FEET	1
	18	EL031-63	CIRCUIT BREAKER 30AMP 480V	1
	19	EL032-47	DOOR DISC OPERATING SHAFT	1
	20	EL032-49	DOOR ROTARY OPERATOR KIT 140G	1
	21	EL032-50	DOOR DISC SWTCH ROTARY W/GUIDE	1
	22	EL040-66	RECEPTICAL DIN RAIL MOUNT 120V	1
	23	EL047-46	TERMINAL BLK. POWER DISTR.	1
	24	EL047-101	TERMINAL BLOCK COVER	1
	25	EL061-26	POWER SUPPLY 380-480AC TO 24VD	1
	26	EL061-27	SURGE SUPPRESSOR 550V 3PHASE	1
	27	EL106-02	RELAY TYPE HC 4PL 4ND/NC 24VDC	1
	A	28	EL107-01	RELAY SOCKET HC 4PL 10A 250V
29		EL109-26	ADJ FREQ DRV AB 20HP 528V 3	1
30		EL152-42	POINT I/O WIRING BASE ASSEMBLY	6
31		EL152-43	POINT I/O 2PORT ETHERNET/IP AD	1
32		EL152-51	POINT I/O 0 ISOLATED THERMOCOUP	2
33		EL152-52	POINT I/O 2 VOLTAGE ANALOG OUT	3
34		TC003-01	DIGITAL READOUT DISPLAY TEMP.	1
35		EL031-66	CIR.BREAKER 10AMP 3POLE AC C	1
36		EL031-69	CIR. BREAKER 6AMP 1POLE AC	1
37		EL031-136	CIRCUIT BREAKER 60 AMP 480V.	1
B	38	EL152-71	POINT I/O 4INPUT ANALOG 4-20MA(NOT SHOWN)	1



REMOVE ALL BURRS AND
BREAK ALL SHARP EDGES
TOLERANCES EXCEPT AS NOTED
DECIMALS .XX ± .01
.XXX ± .005
ANGLES ± 1/2°

**LAWRENCE
EQUIPMENT**

2034 NORTH PECK ROAD
SO. EL MONTE, CALIFORNIA 91733
(626) 442-2894 TELEX 704116
TOLL FREE (800) 423-4500

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CNTRL HEAT EXCHANGE FRYR 460V 0HE3000-02

DRAWN BY
OSCAR MARROQUIN

DATE	7/29/2024
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SCALE:	
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CHECKED BY	
APPROVED BY	

DATE
DATE

DRAWING NO.	REV	SHEET
501667	A	1 of 1

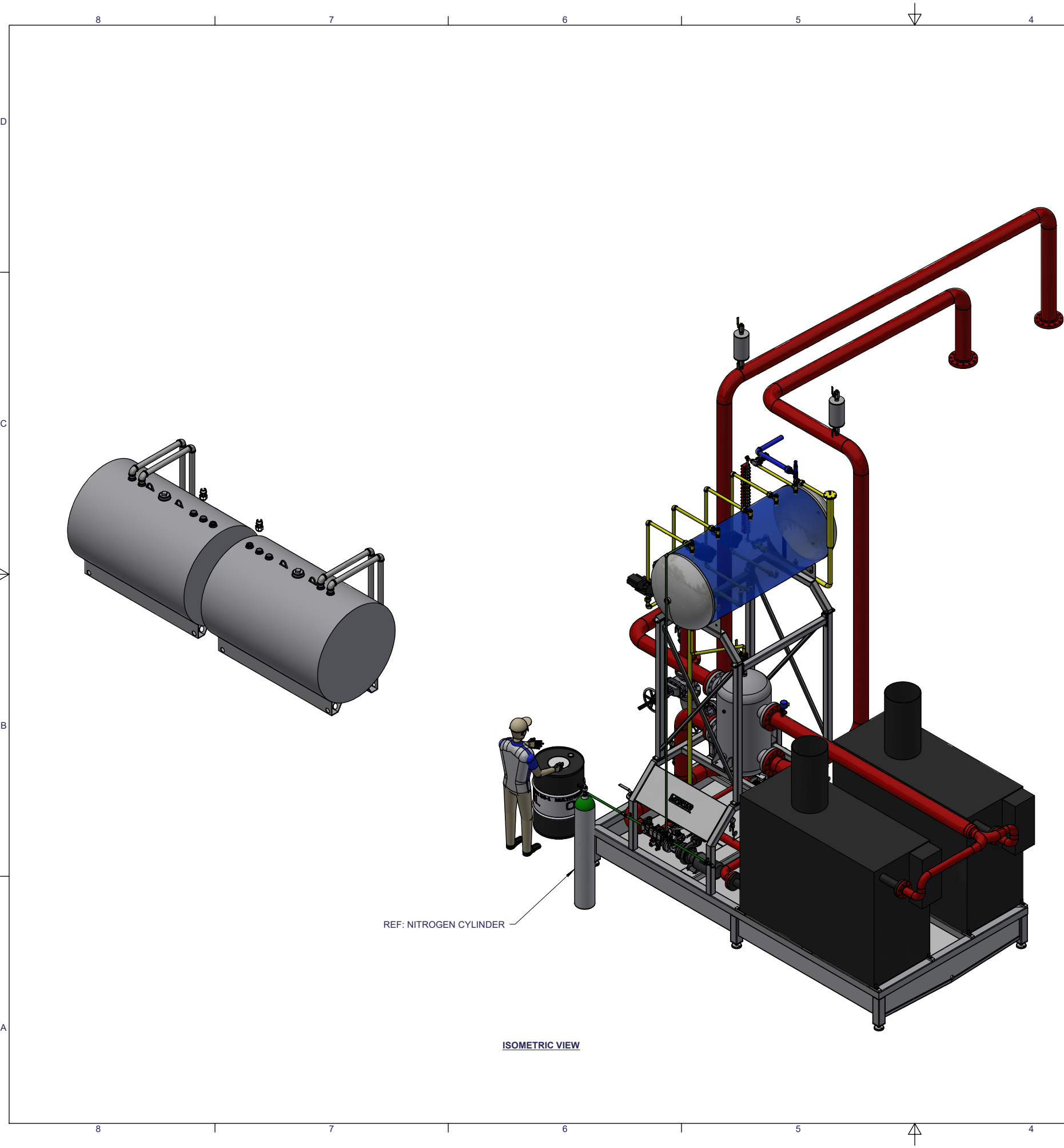
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A	RELEASE TO PRODUCTION	OKM	07/29/24			CHECKED BY	DATE	DRAWING NO. 501667	REV A	SHEET 1 of 1
NO	REVISION	BY	DATE	BY	DATE	APPROVED BY	DATE			



MATERIALS LIST				MATERIALS LIST			
ITEM	LE PART NUMBER	DESCRIPTION	QTY	ITEM	LE PART NUMBER	DESCRIPTION	QTY
60	PF11316	PLUG BLK 2.00 NPT	9	1	242035-001	HEATER BASE STRUCTURE	1
61	PI10840-13.875-TBE	PIPE BLK 1.00 SCHED 40 X 13.88 LG	1	2	242036-001	EXPANSION TANK LEGS	2
62	PI10840-21.75-TBE	PIPE BLK 1.00 SCHED 40 X 21.75 LG	8	3	242037-001	EXPANSION TANK LEGS CROSS MEMB	2
63	PI10840-30.500-TBE	PIPE BLK 1.00 SCHED 40 X 30.50 LG	4	4	280312-001	HEATER INLET THERMO OIL PIPE	2
64	PI11240-12.625-TBE	PIPE BLK 1.50 SCHED 40 X 12.63 LG	2	6	280314-001	HEATER OUTLET THERMO OIL PIPE	1
65	PI11240-14.00-TBE	PIPE BLK 1.50 SCHED 40 X 14.00 LG	2	7	280315-001	NITROGEN PIPE CONNECTION	1
66	PI11240-17.125-TBE	PIPE BLK 1.50 SCHED 40 X 17.13 LG	1	8	280316-001	THERMOFLUID OIL PUMP MANIFOLD	1
67	PI11240-17.5-TBE	PIPE BLK 1.50 SCHED 40 X 17.50 LG	1	9	280317-001	THERMOFLUID OIL PUMP INTAKE	1
68	PI11240-5.250-TBE	PIPE BLK 1.50 SCHED 40 X 5.25 LG	1	10	280318-001	THERMOFLUID OIL RETURN PIPE	1
72	VA002-40	VALVE 1/4NPT CARBON STEEL	5	11	280319-001	THERMOFLUID OIL TANKS CONNECT	1
73	VA002-41	VALVE 1/2NPT CARBON STEEL	1	12	280320-001	THERMOFLUID OIL EXPANSION LINE	1
74	VA002-42	VALVE 1.5NPT CARBON STEEL	3	13	280321-001	THERMOFLUID OIL BREATHING LINE	1
75	VA005-29	CHECK VALVE 0.5NPT 125PSI	1	14	280322-001	THERMOFLUID OIL MAX LEVEL LINE	1
76	VA013-03	SAFETY VALVE	1	15	280323-001	THERMOFLUID OIL MAIN FLOW LINE	1
77	VM054-04	THERMAL FLUID HTR 2.0M BTU	2	16	355794-001	FLOW LEVEL EXTENSION 17.0IN	1
78	VM054-05	THERMAL FLUID EXP. TANK 379GAL	1	17	EL020-57	THERMOCOUPLE ADJ PROBE 3/16	2
79	VM054-06	THERMAL FLUID DIST.TANK 92GAL	1	20	FA260025	REG HEX NUT SS, 1.00-8	6
106	280148-001	AIR COLLECTING TANK 6IN	2	22	HY005-26	GAUGE 60PSI LIQUID FILLED SS	5
107	VA002-47	VALVE 1.0NPT CARBON STEEL	7	23	HY006-20	MULTITHERM OG-1 OPEN CIRCUIT	1
108	PF11308	PLUG BLK, 1.00	2	25	HY012-225	SIGHT GLASS REFLEX TYPE	1
109	AP008-30	NITROGEN BLANKET COMBO ASSY	1	27	MH017-44	HT0180 FLAT FLANGED PUMP 5HP	2
110	PF4090412	NIPPLE SS .50 X 1.5 LONG	2	29	MH020-17	VIBRATION PAD MIG	6
111	PF41104	TEE SS 0.50	2	30	MH031-16	HIGH TEMP LIQUID LEVEL SWITCH	1
112	PI40440	PIPE SS .50 SCHED 40 X 22 in LG.	2	31	MH031-21	THERMAL FLUID LEVEL CUT OFF SW	2
113	PF4010402	BUSHING SS .50 TO .25	1	32	MH031-22	THERMAL FLUID LEVEL CUT OFF SW	2
115	VM054-10	THERMAL FLUID DRAIN TANK 500GA	2	33	MH032-08	THERMOWELL 1/2 NPT SS	2
116	PF1091624	NIPLE BLK Φ 2.00 x 3.00 LG.	4	34	MH040-12	GASKET 150LB FLANGE 1.50	5
117	PF10616	ELBOW BLK Φ 2.00 x 90°	8	35	MH040-13	GASKET 150LB FLANGE 2.00	1
118	PI11640	PIPE BLK 2.00 SCHED 40 X 28.000 in LG.	4	36	MH040-30	METAL REINFORCED 3.0 ANSI300	5
119	PF10316	COUPLING Φ 2.0	4	37	MH040-31	GASKET 150LB FLANGE 3.00	8
120	HY012-246	BREATHER VENT 2IN NPT	4	39	MH040-43	GASKET 3000LB FLANGE 6.00	4
121	PF11332	PLUG BLK 4.00	2	40	MH041-10	GAUGE PROTECTING STEAM 1/4	4
122	PF4011612	BUSHING BLK. 2.00 x 1.500	2	41	PF1010604	BUSHING BLK. 0.75 x 0.50/0.81 OD OUT	1
123	SA001-83	COUPLING S-FLEX FLANGE TP 9SC	2	42	PF1010804	BUSHING BLK 1.00 TO .50	1
124	SA001-82	COUPLING S-FLEX SLEEVE TYPE 9H	1	43	PF1011208	BUSHING BLK 1.50 TO 1.00	2
125	SA001-127	COUPLING S-FLEX HUB 1.125 BORE	1	44	PF1011608	BUSHING BLK. 2.00 x 1.00	1
126	MH017-48	RA3146 3X4X8.5 HOT OIL PUMP	1	45	PF10208	CAP BLK 1.00	3
127	MH040-54	GASKET 300LB FLANGE 4.00 REINF	1	46	PF10602	ELBOW BLK. Φ 0.25 90°	2
129	358796-001	PUMP BASE	1	47	PF10604	ELBOW BLK. Φ 0.50 90°	1
131	MO120-05	MOTOR 20HP 3500RPM 3PH TEFC	1	48	PF10608	ELBOW BLK Φ 1.00 x 90°	11
132	280723-001	PUMP FEEDING ELBOW	1	49	PF10612	ELBOW BLK 1.50	4
133	MH024-15	BELLOWS EXP.JOINT 6INX11 150LB	1	50	PF1090212	NIPPLE BLK. Φ 0.25 x 1.50 LG.	7
134	MH040-34	GASKET 150LB FLANGE 6.00 REINF	3	51	PF1090412	NIPPLE BLK. 0.50 x 1.50 LG.	2
135	SA001-81	COUPLING S-FLEX HUB 1.625 BORE	1	52	PF1090816	NIPPLE BLK Φ 1.00 x 2.00 LG.	29
137	VA002-68	GATE VALVE HIGH TEMP 6.0	1	53	PF1090824	NIPLE BLK Φ 1.00 x 3.00 LG.	6
138	VA002-69	BALL VALVE HIGH TEMP 1.5IN	2	54	PF1091216	NPPLE BLK 1.50X 2.00 LONG	6
139	VA002-63	GATE VALVE HIGH TEMP 3.0	1	55	PF11108	TEE BLK Φ 1.00	7
140	VA002-62	GATE VALVE HIGH TEMP 3.0	2	56	PF11112	TEE BLK 1.50	2
141	MH024-16	BELLOWS EXP.JOINT 3INX6 150LB	2	57	PF11204	UNION BLK 0.50	1
142	PI40440	PIPE SS .50 SCHED 40 X 12"	1	58	PF11208	UNION BLK 1.0	11
143	PI11640	PIPE BLK 2.00 SCHED 40 X 45.000 in LG.	4	59	PF11212	UNION BLK 1.50	4

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RELEASE TO PRODUCTION

OKM

08/21/24

NO

REVISION

BY

DATE

BY

DATE

REMOVE ALL BURRS AND
BREAK ALL SHARP EDGES
TOLERANCES EXCEPT AS NOTED
DECIMALS .XX ± .01
XXX ± .005
ANGLES ± 1/2°

LAWRENCE
EQUIPMENT

2034 NORTH PECK ROAD
SO. EL MONTE, CALIFORNIA 91733
(626) 442-3894 TELEX 704116
TOLL FREE (800) 423-4500

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PERMISSION OF THE OWNER.

THERMOFLUID HEATER ASSY 4.0MM

OCF6018-01

DRAWN BY
Nicola

DATE
10/10/2014

CHECKED BY

DATE

APPROVED BY

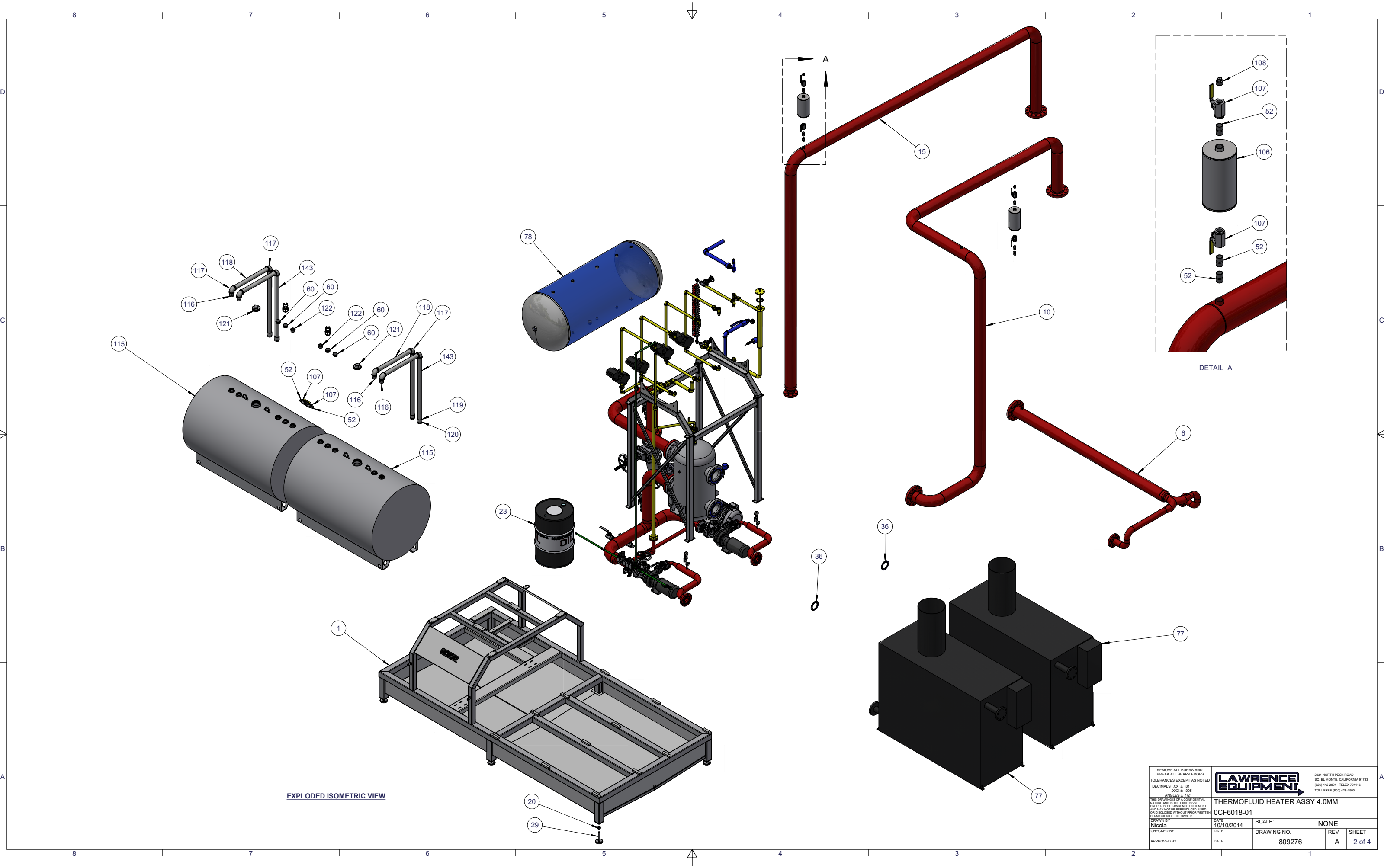
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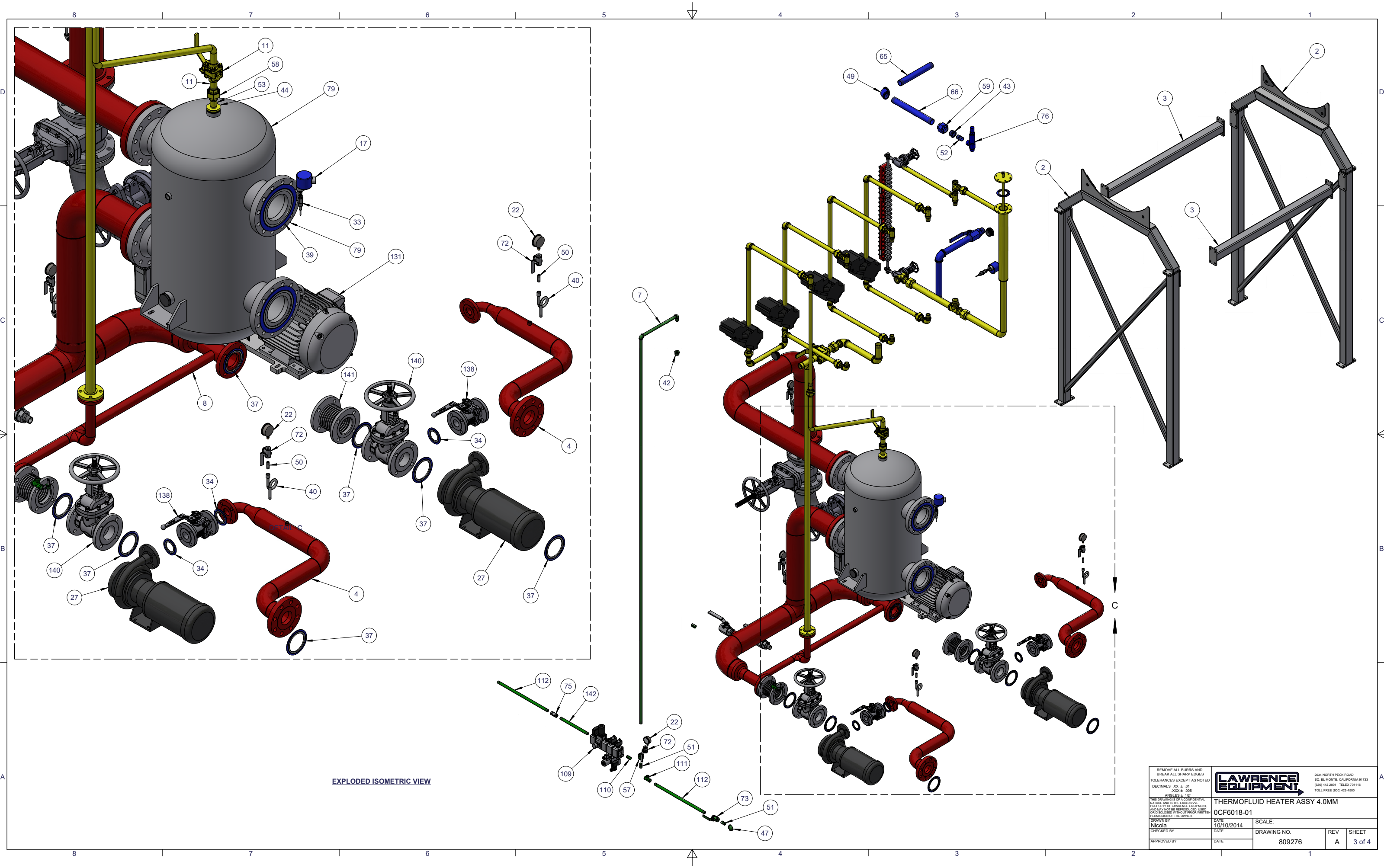
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1 of 4



EXPLODED ISOMETRIC VIEW

REMOVE ALL BURRS AND BREAK ALL SHARP EDGES TOLERANCES EXCEPT AS NOTED DECIMALS .XX ± .01 XXX ± .005 ANGLES ± 1°		LAWRENCE EQUIPMENT		2034 NORTH PECK ROAD SO. EL MONTE, CALIFORNIA 91733 (626) 442-3884 TELEFAX 704116 TOLL FREE (800) 423-4500	
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Ocf6018-01					
DRAWN BY Nicola	DATE 10/10/2014	SCALE: NONE			
CHECKED BY	DATE	DRAWING NO. 809276	REV A	SHEET 2 of 4	
APPROVED BY	DATE				



EXPLODED ISOMETRIC VIEW

REMOVE ALL BURRS AND BREAK ALL SHARP EDGES TOLERANCES EXCEPT AS NOTED DECIMALS .XX ± .01 XXX ± .005 ANGLES ± 1°		LAWRENCE EQUIPMENT		2034 NORTH PECK ROAD SO. EL MONTE, CALIFORNIA 91733 (626) 442-2884 TELEX 704116 TOLL FREE (800) 423-4500	
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DRAWN BY Nicola	DATE 10/10/2014	SCALE:		REV	SHEET
CHECKED BY	DATE	DRAWING NO. 809276		A	3 of 4
APPROVED BY	DATE				

